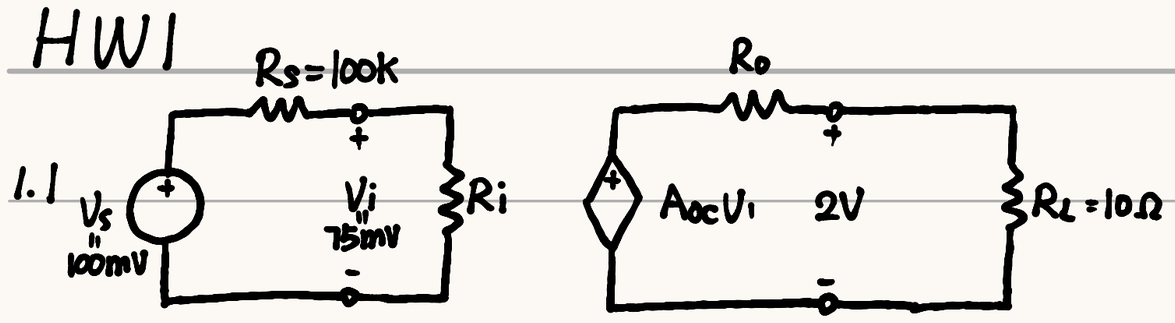
HW1

1.1

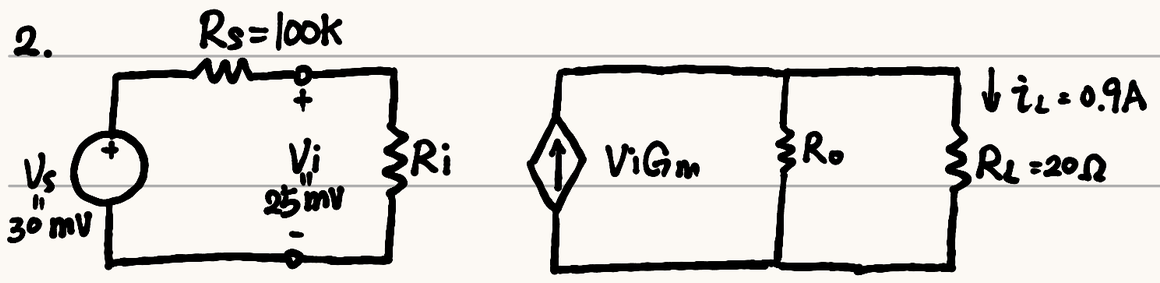


$$V_i = V_s \frac{R_i}{R_i+R_s} = 100m \frac{R_i}{R_i+100k} = 75mV \Rightarrow R_i = 300k\Omega$$

$$\begin{cases} V_o = A_{oc}V_i = 75mA_{oc}\dfrac{10}{10+R_o} = 2 \\ V_o' = 75mA_{oc}\dfrac{10//30}{(10//30)+R_o} = 75mA_{oc}\dfrac{7.5}{7.5+R_o} = 1.8 \end{cases} \Rightarrow \begin{cases} A_{oc}=40 \\ R_o=5 \end{cases}$$

$\therefore R_i = 300k\Omega \quad A_{oc} = 40 \quad R_o = 5\Omega$

2.



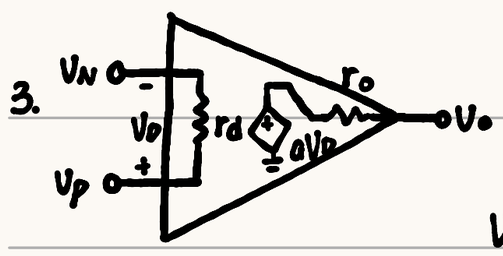
$$\begin{cases} 25mVGm \times \dfrac{R_o}{R_o+20} = 0.9 \\ 25mVGm \times \dfrac{R_o}{R_o+30} = 0.8 \end{cases} \Rightarrow \begin{cases} Gm = 48 \\ R_o = 60 \end{cases}$$

$$V_i = V_s \frac{R_i}{R_s+R_i} = 30mV \times \frac{R_i}{100k+R_i} = 25mV \Rightarrow R_i = 500k$$

$$V_i' = 33mV \frac{500k}{50k+500k} = 30mV$$

$$i_L = 30mV \times 48 \frac{60}{60+40} = 0.864A$$

3.



$$V_o = a(V_p - V_N)$$

(a) $V_o = 10^4(750.25 - 751.50)mV = -12.5V$

(b) $-5 = 10^4(0 - V_N) \Rightarrow V_N = 0.5mV$

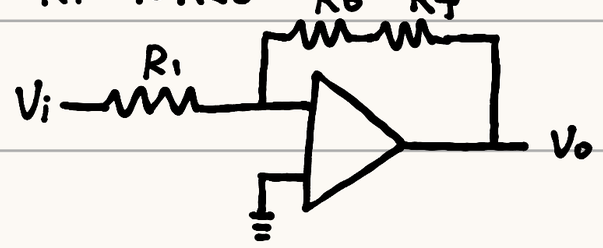
(c) $5 = 10^4(V_p - 5) \Rightarrow V_p = 5.0005V$

(d) $-1 = 10^4(1 - V_N) = 1.0001V$

4. (a) $A = -\dfrac{R_f}{R_1} \quad A \in [-10, 0] \quad R_f \in [0, 100k\Omega]$

$\Rightarrow R_1 = 10k\Omega$

(b)



$$A = -\frac{R_o+R_f}{R_1}$$

$$A = -\frac{R_o+R_f}{R_1} \in \left[-\frac{R_o+100k}{R_1}, \frac{R_o}{R_1}\right]$$

$$\begin{cases} \dfrac{R_o+100k}{R_1} = 10 \\ \dfrac{R_o}{R_1} = 1 \end{cases} \Rightarrow \begin{cases} R_o = \dfrac{100}{9}k \approx 11.11k\Omega \\ R_1 \approx 11.11k\Omega \end{cases}$$

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